



RESEARCH ARTICLE

Role of integrated nutrient management in enhancing quality of mungbean (*Vigna radiata* L.) varieties

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Abstract

The field experiment was conducted at the Crop Research Farm, Faculty of Agriculture, Maharishi Markandeshwar (Deemed to be) University, Mullana, Haryana during the 2024 and 2025 summer seasons, to evaluate the effect of integrated nutrient management (INM) on nutrient content and uptake of mungbean (*Vigna radiata* L.). The experiment was conducted in a factorial randomised block design (RBD) with 3 replications, comprising 3 mungbean varieties (MH 421, MH 1142 and MH 1762) and eight nutrient management treatments (T_1 – T_8), including graded nitrogen (N) and phosphorus (P) levels, *Rhizobium* inoculation and recommended dose of fertilisers (RDF) integrated with nano urea, nano diammonium phosphate (DAP) and *Rhizobium*. The results revealed significant variation among varieties and nutrient management treatments. Among the varieties, MH 1142 consistently recorded higher N content in seed that was 5.28 in 2024 and 23.22 in 2024 and P content in seed was 2.42 in 2024 and 15.24 in 2024 and 15.29 in 2025. Among INM treatments, application of 20 kg N + 60 kg phosphorus pentoxide (P_2O_5) along with *Rhizobium* inoculation (T_6) significantly enhanced nutrient content and uptake, owing to improved nutrient availability and biological N fixation. Nano-fertiliser based treatments showed promising effects, particularly in improving P content, but were slightly inferior to conventional fertiliser combined with *Rhizobium* inoculation. The MH 1142 recorded the highest N and P content as well as uptake in both seed and straw during both years, with N content of 5.28 and 5.32 % in seed, 3.32 and 3.36 % in straw, P content of 2.42 and 2.45 % in seed and 1.54 and 1.56 % in straw during 2024 and 2025, respectively. Similarly, it registered the maximum N uptake in seed (23.22 and 23.68 kg ha⁻¹) and straw (12.93 and 13.02 kg ha⁻¹) and P uptake in seed (15.24 and 15.29 kg ha⁻¹) and straw (7.25 and 7.29 kg ha⁻¹), indicating its superior nutrient acquisition and utilisation efficiency.

Keywords: integrated nutrient management; mungbean; nano-fertilisers; nutrient uptake; *Rhizobium*

Introduction

Mungbean (*Vigna radiata* (L.) is one of the predominant grain legume crops of South and Southeast Asia and holds a prominent place in Indian agriculture due to its short duration, adaptability to diverse agro-ecological conditions and high nutritional value. It is popularly known as the “Golden Bean” because of its rich protein content (22–26 %), appreciable carbohydrate concentration (approximately 60 %) and significant amounts of important minerals such as phosphorus (P), calcium (Ca), iron (Fe) and potassium (K). Mung bean is a nutrient-dense pulse with an average protein content of 23.8 g/100 g dry matter (DM) and carbohydrate content of 61.0 g/100 g DM, making it an excellent source of plant-based protein and energy. It contains low fat (1.22 g/100 g DM), moderate crude fibre (4.57 g/100 g DM) and ash (3.51 g/100 g DM), with an average moisture content of 9.80 g/100 g. The average energy value of mung bean is 344 kcal/100 g DM, highlighting its high nutritional value (1, 2).

With the rapid increase in global population, ensuring food and nutritional security while conserving natural resources has become a major challenge. Intensive agricultural practices relying heavily on chemical fertilisers have undoubtedly enhanced crop

productivity, increasing seed yield by 20–50 % but have simultaneously resulted in declining nutrient use efficiency (NUE) with N recovery rarely exceeding 30–50 % and P recovery only 10–25 % during the first cropping season, leading to considerable nutrient losses and adverse environmental impacts, soil degradation, nutrient imbalances and environmental pollution. Excessive or imbalanced fertiliser application has been associated with reduced soil organic carbon (SOC), nutrient mining, groundwater contamination and increased greenhouse gas (GHG) emissions, thereby threatening long-term agricultural sustainability (3). Consequently, sustainable nutrient management strategies that enhance productivity without compromising soil health are urgently required.

Integrated nutrient management (INM), which consist of the combined use of chemical fertilisers, organic manures and biofertilisers, has emerged as a practical strategy to maintain soil fertility, improve nutrient availability and enhance crop quality. Integrated nutrient management ensures synchronised nutrient release, improved NUE and sustained soil biological activity, thereby promoting sustainable crop production. Recent studies have demonstrated that integrated application of organic sources along with suggested doses of inorganic fertilisers significantly

improve soil physicochemical properties, microbial activity and nutrient availability compared to sole application of chemical fertilisers (4, 5).

Nitrogen and P are two most critical nutrients influencing growth, yield and seed quality of mungbean. Nitrogen is an important element of amino acids, proteins, nucleic acids and chlorophyll and its adequate availability directly determines seed protein content and nutritional quality (6). Phosphorus plays a key role in root development, nodulation, energy transfer, photosynthesis and seed formation. However, P use efficiency in agricultural soils remains low, as merely 25–30 % of the applied dose, P is utilised by crops, whereas the remaining portion becomes fixed in non-soluble forms due to interactions with Ca, Fe, aluminium (Al) and magnesium (Mg) in the soil matrix (7, 8). This low availability of P often limits crop productivity and nutrient uptake, particularly in alkaline and carbonate-rich soils.

Nano-fertilisers have emerged as an innovative tool for improving fertiliser-use efficiency through controlled nutrient release, enhanced nutrient uptake and reduced environmental losses (9). In mungbean, nano urea and nano diammonium phosphate (DAP) improve N and P metabolism, photosynthesis, nodulation and biological N fixation, resulting in better growth, yield and seed quality. Similarly, *Rhizobium* inoculation enhances symbiotic N fixation, nutrient availability, root development and the production of plant growth-promoting substances, while improving soil microbial activity and fertility (10). Therefore, integrating nano-fertilisers with *Rhizobium* and conventional fertilisers offer a sustainable nutrient management strategy for enhancing productivity and maintaining soil health in mungbean (11, 12).

Recent field studies have demonstrated that the integrated application of organic manures, inorganic fertilisers and biofertilisers significantly increase N and P concentration and uptake in mungbean seed and straw, resulting in improved seed protein content and overall seed quality (12, 13). This INM approach ensures a continuous supply of nutrients, particularly N, during critical crop growth stages, thereby enhancing protein synthesis, NUE and crop productivity (14).

Although several studies have demonstrated the individual benefits of inorganic fertilisers, *Rhizobium* inoculation and nano-fertilisers in mungbean, limited information is available on their integrated application under different nutrient management regimes. Moreover, comparative information on the response of different mungbean varieties to the combined use of inorganic fertilisers, *Rhizobium* and nano-fertilisers, particularly with respect to nutrient concentration, nutrient uptake and seed protein content under the agro-climatic conditions of Haryana, is scarce. This lack of region-specific information restricts the development of efficient and sustainable nutrient management strategies for maximising productivity and seed quality. Therefore, the present investigation was undertaken to evaluate the effects of INM practices involving inorganic fertilisers, *Rhizobium* inoculation and nano-fertilisers on nutrient content, nutrient uptake and protein status of different mungbean (*V. radiata*) varieties under the prevailing agro-climatic conditions of Haryana. The study also aimed to identify the most suitable variety and nutrient management combination for improving nutrient uptake, seed quality and sustainable crop productivity.

Materials and Methods

Experimental location and soil characteristics

The experiment was conducted during two consecutive summer cropping period of 2024 and 2025 at the Crop Research Farm, Faculty of Agriculture, Maharishi Markandeshwar (Deemed to be) University, Mullana, Ambala, Haryana, India. The experimental site is located at the specified geographical conditions that is at the latitude of 30°17'0" N and longitude of 77°03'0" E, at an elevation of 264 m AMSL. Soil analysis of the experimental site was carried out to determine its physico-chemical properties (Table 1). Composite soil samples (0–15 cm depth) were collected before sowing and analysed using standard procedures. Soil texture was determined by the Bouyoucos hydrometer method (15); soil pH and electrical conductivity (EC) were measured in a 1 : 2 soil-water suspension using a digital pH meter and conductivity meter, respectively. Organic carbon was estimated by the Walkley and Black wet oxidation method (15), available N by the alkaline permanganate method (16), available P by Olsen's method (17) and available K by the neutral normal ammonium acetate (NH₄CH₃COO) extraction method followed by flame photometry. The soil was sandy clay loam in texture, comprising 54.40 % sand, 28.60 % silt and 17.00 % clay. The higher proportion of sand suggested good drainage and aeration, while the adequate silt and clay fractions enhanced moisture and nutrient retention, making the soil suitable for mungbean cultivation due to favourable root development and nodulation.

Table 1. Physical and chemical properties of soil of the experimental field

Physical analysis of the soil of experimental field	
Soil properties	Value (%)
Sand	54.40
Silt	28.60
Clay	17.00
Chemical analysis of the soil of experimental field	
Soil pH (1: 2.5)	7.98
EC (dS m ⁻¹)	0.89
Organic C (%)	0.31
Available N (kg ha ⁻¹)	173
Available P ₂ O ₅ (kg ha ⁻¹)	9.96
Available K ₂ O (kg ha ⁻¹)	470

EC: electrical conductivity; N: nitrogen; C: carbon.

The chemical analysis revealed that the soil was slightly alkaline (pH 7.98). The EC (0.89 dS m⁻¹) indicated that the soil was non-saline and suitable for normal crop growth. The organic C content was low (0.31 %), indicating poor organic matter status and limited native fertility. The available N (173 kg ha⁻¹) and P (9.96 kg P₂O₅ ha⁻¹) contents were low, suggesting the need for external nutrient supplementation for optimum crop growth, root development and nodulation. In contrast, the soil had a high available K content (470 kg K₂O ha⁻¹), indicating an adequate K supply for mungbean cultivation.

Overall, the soil fertility status justified the application of INM practices involving chemical fertilisers, biofertilisers and nano-fertilisers to improve nutrient availability, uptake and productivity of mungbean.

Experimental design

The research trial was conducted in a factorial randomised block design (RBD) with 3 replications. The treatment structure comprises of three green gram (*V. radiata*) varieties, namely MH421, MH1142 and MH1762 and 8 INM treatments consisted of chemical fertilisers, biofertilisers and nano-fertilisers.

The nutrient management treatments were as follows: T₁: 15 kg N + 40 kg P₂O₅ ha⁻¹; T₂: 15 kg N + 60 kg P₂O₅ ha⁻¹; T₃: 20 kg N + 60 kg P₂O₅ ha⁻¹; T₄: 15 kg N + 40 kg P₂O₅ ha⁻¹ + *Rhizobium*; T₅: 15 kg N + 60 kg P₂O₅ ha⁻¹ + *Rhizobium*; T₆: 20 kg N + 60 kg P₂O₅ ha⁻¹ + *Rhizobium*; T₇: recommended dose of fertilisers (RDF) + 2 mL nano urea + nano DAP ha⁻¹ + *Rhizobium* and T₈: RDF + 4 mL nano urea + nano DAP ha⁻¹ + *Rhizobium*. Each experimental plot measured 3 × 2 m and the RDF for mungbean is 20: 40: 20 kg ha⁻¹ of N : P : K.

Seed inoculation and sowing

The seeds of mungbean varieties (MH 421, MH 1142 and MH 1762) were procured from the department seed store and sown at the rate of 25 kg ha⁻¹. For *Rhizobium* inoculation, seeds were divided into required portions and placed in separate polythene bags. The *Mesorhizobium* strain of *Rhizobium* inoculum applied at the rate of 25 g kg⁻¹ of seed, followed by mixing to ensure uniform coating. Care was taken to avoid cross-contamination among treatments. After inoculation, seeds were shade-dried and immediately sown in the field.

Crop management practices

Fertilisers (urea, DAP, nano urea and nano DAP) were incorporated according to the respective treatment combinations at the time of sowing. Nano urea and nano DAP were applied as foliar sprays at concentrations of 2 and 4 mL L⁻¹ of water. Nano DAP was sprayed once at the active vegetative stage (30–35 days after sowing (DAS)), whereas nano urea was applied once at the pre-flowering stage (40–45 DAS) using a knapsack sprayer to ensure uniform coverage of the foliage. The foliar sprays were carried out during the morning hours under calm weather conditions to maximise nutrient absorption and minimise spray losses.

Thinning was carried out at 15 DAS to maintain a plant-to-plant spacing of 10 cm and that of row-to-row spacing of 30 cm. Weed management was carried out manually through hand weeding at 30 and 60 DAS to maintain a weed-free crop environment. Irrigation and plant protection measures were adopted as and when required to maintain optimum crop growth. At physiological maturity, the crop was harvested and observations on grain and stover yields were documented.

Data collection

Observations on quality parameters were recorded after the harvest of crop. Five representative plants were randomly selected and tagged in each plot for recording quality attributes. Quality

parameters were assessed by estimating nutrient (N and P) content in grain and nutrient uptake was calculated using Micro-Kjeldahl method following the procedures of Association of Official Analytical Chemists (AOAC) (18) for N. Phosphorus content was estimated spectrophotometrically by the vanadomolybdate yellow colour method after wet acid digestion of the seed samples as described by AOAC.

Nutrient uptake by seed was calculated by multiplying the nutrient concentration (%) with the corresponding seed yield (kg ha⁻¹) and dividing by 100, using the following formula:

Nutrient uptake (kg ha⁻¹)

$$= [\text{Nutrient content (\%)} \times \text{Seed yield (kg ha}^{-1}\text{)}] / 100$$

Statistical analysis

The experimental data recorded for various parameters were subjected to statistical analysis using OPSTAT, an online statistical analysis software developed by CCS Haryana Agricultural University, Hisar, for the analysis of variance (ANOVA) appropriate to a factorial RBD (19). Treatment effects were evaluated through the 'F' test. The standard error of mean (SEm) was computed and significant differences among treatment means were determined using the critical difference (CD) test at the 5 % level of significance.

The CD was calculated using the formula:

$$CD = \sqrt{\frac{2 \times \text{error sum of square}}{N}} \times t(\text{error d.f } 5 \%)$$

Results and Discussion

The findings presented in the Table 2, revealed that INM significantly influenced plant height, number of seeds per plant and seed yield of mungbean, while varietal differences significantly affected plant height and number of seeds per plant. However, seed yield among varieties did not differ significantly in either year of the study (Fig. 1).

Among the varieties, V₂ recorded the maximum plant height (72.32 and 73.57 cm during 2024 and 2025, respectively), which was significantly superior to V₁ and V₃. The higher plant height in V₂ may be attributed to its better genetic potential, vigorous vegetative growth and greater nutrient utilisation efficiency. The number of seeds per plant was also highest in V₂ (4.85 and 6.00), followed closely by V₁, whereas the minimum values were observed in V₃. Although V₂ produced the highest seed yield (996.52 and 1015.85 kg ha⁻¹), the differences among varieties were statistically

Table 2. Effect of integrated nutrient management (INM) on the plant height, number of seeds per plant and seed yield of different varieties of green gram

Treatments	Plant height (cm)		No. of seeds per plant		Seed yield kg/ha	
	2024	2025	2024	2025	2024	2025
Varieties						
V ₁	61.45	62.54	4.87	5.96	968.52	993.52
V ₂	72.32	73.57	4.85	6.00	996.52	1015.85
V ₃	64.59	64.87	4.44	5.62	964.03	972.82
SEm (±)	0.76	0.64	0.11	0.10	15.05	14.01
CD (at 5 %)	1.53	1.29	0.21	0.21	NS	NS
Integrated nutrient management						
T ₁	61.99	60.91	3.61	4.62	911.95	914.95
T ₂	62.89	63.44	3.63	4.68	926.78	937.44
T ₃	67.71	68.96	5.63	6.81	1013.39	1034.72
T ₄	64.90	66.15	4.26	5.44	953.07	974.41
T ₅	68.46	69.71	5.64	6.82	1017.22	1038.55
T ₆	72.04	73.29	6.06	7.23	1068.65	1089.88
T ₇	65.47	66.72	4.34	5.52	956.15	977.48
T ₈	65.51	66.76	4.59	5.77	963.75	985.08
SEm (±)	1.24	1.05	0.17	0.17	24.57	22.87
CD (at 5 %)	2.50	2.11	0.35	0.34	49.46	46.04

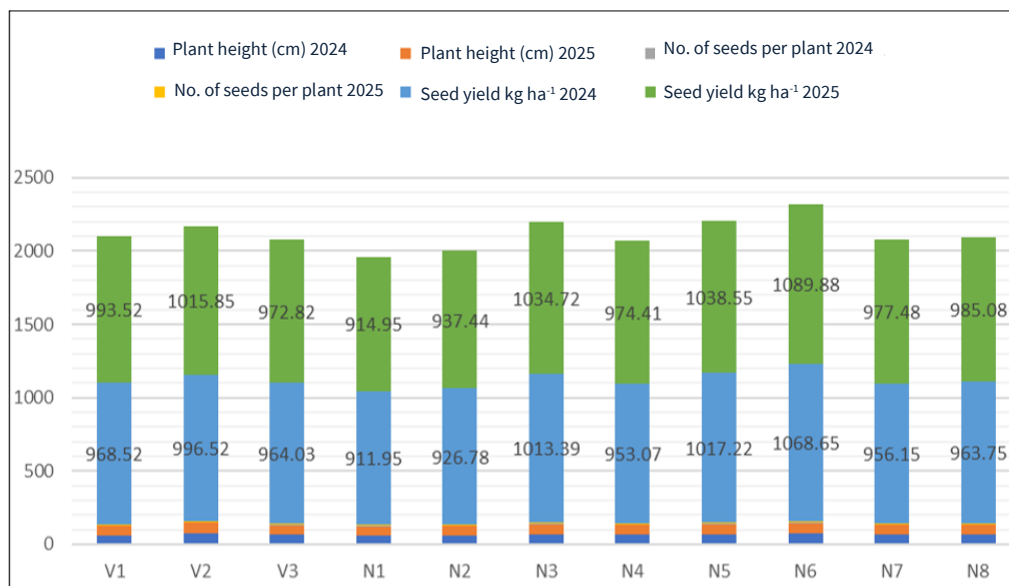


Fig. 1. Effect of integrated nutrient management (INM) on the plant height and dry matter accumulation of different varieties of green gram.

non-significant, indicating similar yield performance under the prevailing conditions.

Integrated nutrient management treatments exerted a substantial influence on growth and yield performance. Among the different treatments, T₆ (20 kg of N + 60 kg of P₂O₅ + *Rhizobium*) recorded the maximum plant height (72.04 and 73.29 cm), number of seeds per plant (6.06 and 7.23) and seed yield (1068.65 and 1089.88 kg ha⁻¹) during 2024 and 2025, respectively.

The superior performance of T₆ may be attributed to the synergistic effects of balanced fertilisation and *Rhizobium* inoculation, which enhanced nutrient availability, NUE and biological N fixation throughout the crop growth period. The adequate supply of N and P during the early growth stages promoted vigorous root development, effective nodulation, chlorophyll synthesis, ATP production and enhanced photosynthetic activity, while *Rhizobium* inoculation ensured a sustained N supply during the reproductive phase. Improved nutrient acquisition and metabolic activity facilitated greater dry matter accumulation, efficient assimilate translocation from source to sink, better flower retention, pod setting and seed filling, ultimately resulting in higher yield attributes and seed yield.

Similarly, treatments T₅ and T₃ also recorded comparatively higher values for yield attributes and seed yield owing to the balanced supply of N and P, which improved enzymatic activity, carbohydrate synthesis and partitioning and overall physiological efficiency, thereby supporting enhanced reproductive development and crop productivity.

The lowest plant height, seeds per plant and seed yield were observed under T₁ (15 kg N + 40 kg P₂O₅), indicating that lower fertiliser doses without biofertiliser inoculation were insufficient to ensure an adequate nutrient supply for crop growth. Nano-fertiliser treatments (T₇ and T₈) improved growth and yield parameters compared to lower fertiliser levels, reflecting the beneficial role of nano nutrients in improving NUE; however, their performance remained slightly inferior to T₆. Overall, the findings indicated that integration of inorganic nutrient sources with *Rhizobium* inoculation resulted in a significant enhancement in mungbean growth and yield (20–21).

Nitrogen content in the seed varied significantly (Table 3) among varieties and nutrient management treatments during both years. Among varieties, MH 1142 (V₂) recorded the highest N content

Table 3. Influence of integrated nutrient management (INM) and varieties on nitrogen and phosphorus content in seed and straw of mungbean (*Vigna radiata* L.)

Treatments	Nitrogen content in seed (%)		Nitrogen content in straw (%)		Phosphorus content in seed (%)		Phosphorus content in straw (%)	
	2024	2025	2024	2025	2024	2025	2024	2025
Varieties								
V ₁ (MH 421)	5.17	5.19	3.26	3.28	2.38	3.39	1.45	1.47
V ₂ (MH 1142)	5.28	5.32	3.32	3.36	2.42	2.45	1.54	1.56
V ₃ (MH 1762)	4.20	4.21	2.29	2.32	1.48	1.49	0.90	0.91
SEm (±)	0.11	0.12	0.18	0.16	0.17	0.19	0.15	0.17
CD (at 5 %)	0.26	0.25	0.35	0.36	0.34	0.28	0.29	0.23
Integrated nutrient management								
T ₁ (15 kg N + 40 kg P ₂ O ₅)	2.26	2.29	1.17	1.19	0.72	0.74	0.29	0.31
T ₂ (15 kg N + 60 kg P ₂ O ₅)	2.57	2.61	1.14	1.21	0.91	0.95	0.54	0.59
T ₃ (20 kg N + 60 kg P ₂ O ₅)	3.09	3.12	1.86	1.93	1.30	1.35	0.74	0.76
T ₄ (15 kg N + 40 kg P ₂ O ₅ + <i>Rhizobium</i>)	3.02	3.08	1.79	1.82	1.23	1.24	0.70	0.73
T ₅ (15 kg N + 60 kg P ₂ O ₅ + <i>Rhizobium</i>)	2.77	2.82	1.54	1.58	0.98	1.02	0.51	0.53
T ₆ (20 kg N + 60 kg P ₂ O ₅ + <i>Rhizobium</i>)	3.94	3.99	2.71	2.76	1.32	1.35	1.59	1.61
T ₇ (RDF + 2 mL nano urea + 2 mL nano DAP ha ⁻¹)	3.44	3.48	2.21	2.25	1.6	1.62	1.17	1.19
T ₈ (RDF + 4 mL nano urea + 4 mL nano DAP ha ⁻¹)	3.31	3.39	2.08	2.18	1.5	1.51	0.96	1.08
SEm (±)	0.16	0.15	0.29	0.21	0.28	0.25	0.24	0.26
CD (at 5 %)	0.25	0.33	0.58	0.51	0.55	0.44	0.48	0.49

DAP: diammonium phosphate; RDF: recommended dose of fertilisers.

in seed (5.28 % in 2024 and 5.32 % in 2025), which was statistically superior to MH 1762 (V₃). The minimum N content was observed in MH 1762 (V₃) (4.20 and 4.21 %). The higher seed N in MH 1142 may be associated with the greater N uptake capacity and efficient translocation of assimilates towards developing grains. Among INM treatments, T₆ (20 kg N + 60 kg P₂O₅ + *Rhizobium*) recorded the highest N concentration in seed (3.94 and 3.99 %), whereas T₁ (15 kg N + 40 kg P₂O₅) led to the lowest values (2.26 and 2.29 %). The superiority of T₆ can be attributed to higher N availability coupled with biological N fixation by *Rhizobium*, leading to improved N assimilation during seed development (22, 23).

Varietal differences and nutrient management practices significantly affected the N concentration in straw. Among varieties, MH 1142 (V₂) recorded the maximum straw N content (3.32 and 3.36 %), while MH 1762 (V₃) registered the minimum values (2.29 and 2.32 %). This indicates a greater overall N uptake and retention in vegetative tissues in V₂. Among INM treatments, T₆ recorded the highest N content in straw (2.71 and 2.76 %), whereas T₁ resulted in the lowest N content (1.17 and 1.19 %).

The higher N content in straw of MH 1142 could be attributed to its enhanced vegetative growth, better root proliferation and superior nutrient absorption efficiency, which enabled greater accumulation of nitrogenous compounds in plant tissues. In addition, efficient photosynthetic activity and higher biomass production in this variety may have contributed to improved synthesis and storage of proteins and amino acids in the straw. The combined application of higher N and P levels along with *Rhizobium* inoculation might have improved nodulation and biological N fixation, thereby increasing N availability to the crop. Adequate P supply also plays a crucial role in energy transfer and root development, which enhances nutrient uptake efficiency. Furthermore, *Rhizobium* inoculation may have stimulated the production of growth-promoting substances, resulting in more extensive root system and greater acquisition of nutrients from the soil (24). The synergistic use of fertilisers and biofertilisers facilitated

balanced nutrient availability and efficient N metabolism, leading to higher N accumulation in straw (25, 26).

Phosphorus content in seed differed significantly among varieties and treatments during both seasons. The MH 1142 (V₂) recorded the highest P content in seed (2.42 and 2.45 %), while MH 1762 (V₃) showed the lowest values (1.48 and 1.49 %). Higher P content in MH 1142 may be attributed to better P uptake efficiency and effective utilisation for energy transfer and seed formation. Among nutrient management treatments, T₇ (RDF + nano urea + nano DAP ha⁻¹) resulted in the maximum P content in seed (1.60 and 1.62 %), whereas T₁ recorded the minimum values (0.72 and 0.74 %). The enhanced P content under nano-DAP treatments may be due to higher solubility, rapid absorption and improved P use efficiency (27–28).

As data represented in the Table 4, it indicates that, among the varieties, V₂ (MH 1142) observed the highest nitrogen uptake in seed during both years (23.22 and 23.68 kg ha⁻¹ in 2024 and 2025, respectively), which was significantly superior to V₃ and at par with V₁. The minimum N uptake was observed in V₃ (MH 1762) (22.13 and 23.12 kg ha⁻¹ in 2024 and 2025) (Fig. 2).

The higher N uptake in V₂ may be attributed to its superior genetic potential for nutrient acquisition and utilisation, which enabled the development of a more efficient root system for greater N absorption from the soil. In addition, V₂ exhibited better nodulation and biological N fixation, enhanced photosynthetic efficiency and greater dry matter production, resulting in improved nutrient translocation and partitioning towards developing seeds. Its comparatively higher seed yield also contributed to increased N uptake, as nutrient uptake is directly influenced by both nutrient concentration and biomass production. Conversely, the lower N uptake recorded in V₃ may be associated with its relatively lower nutrient acquisition efficiency, reduced biomass accumulation and lower seed yield, thereby limiting the total amount of N absorbed and translocated to the seeds.

Table 4. Influence of integrated nutrient management (INM) and varieties on nitrogen (N) and phosphorus (P) uptake in

Treatments	Nitrogen uptake in seed (kg ha ⁻¹)		Nitrogen uptake in straw (kg ha ⁻¹)		Phosphorus uptake in seed (kg/ha)		Phosphorus uptake in straw (kg/ha)	
	2024	2025	2024	2025	2024	2025	2024	2025
Varieties								
V ₁ (MH 421)	23.10	23.52	12.81	12.99	15.12	15.21	7.13	7.21
V ₂ (MH 1142)	23.22	23.68	12.93	13.02	15.24	15.29	7.25	7.29
V ₃ (MH 1762)	22.13	23.12	11.84	11.92	14.15	14.50	6.16	6.19
SEm (±)	0.19	0.12	0.19	0.21	0.22	0.31	0.22	0.34
CD (at 5 %)	0.39	0.36	0.39	0.35	0.41	0.32	0.41	0.25
Integrated nutrient management								
T ₁ (15 kg N + 40 kg P ₂ O ₅)	13.47	13.56	7.04	7.12	8.48	8.56	3.49	3.52
T ₂ (15 kg N + 60 kg P ₂ O ₅)	13.78	13.88	7.35	7.38	8.79	8.82	3.80	3.92
T ₃ (20 kg N + 60 kg P ₂ O ₅)	14.30	14.36	7.87	7.91	9.31	9.36	4.32	4.44
T ₄ (15 kg N + 40 kg P ₂ O ₅ + <i>Rhizobium</i>)	14.23	14.32	7.80	7.99	9.24	9.33	4.25	4.32
T ₅ (15 kg N + 60 kg P ₂ O ₅ + <i>Rhizobium</i>)	13.98	14.01	7.55	7.62	8.99	9.23	4.00	4.22
T ₆ (20 kg N + 60 kg P ₂ O ₅ + <i>Rhizobium</i>)	15.15	15.24	8.72	8.85	10.16	10.19	5.17	5.21
T ₇ (RDF + 2 mL nano urea + 2 mL nano DAP ha ⁻¹)	14.65	14.76	8.22	8.35	9.66	9.72	4.67	4.69
T ₈ (RDF + 4 mL nano urea + 4 mL nano DAP ha ⁻¹)	14.52	14.66	8.09	8.15	9.53	9.65	4.54	4.61
SEm (±)	0.32	0.31	0.33	0.22	0.26	0.29	0.58	0.55
CD (at 5 %)	0.64	0.53	0.60	0.69	0.58	0.46	0.21	0.45

DAP: diammonium phosphate; RDF: recommended dose of fertilisers.

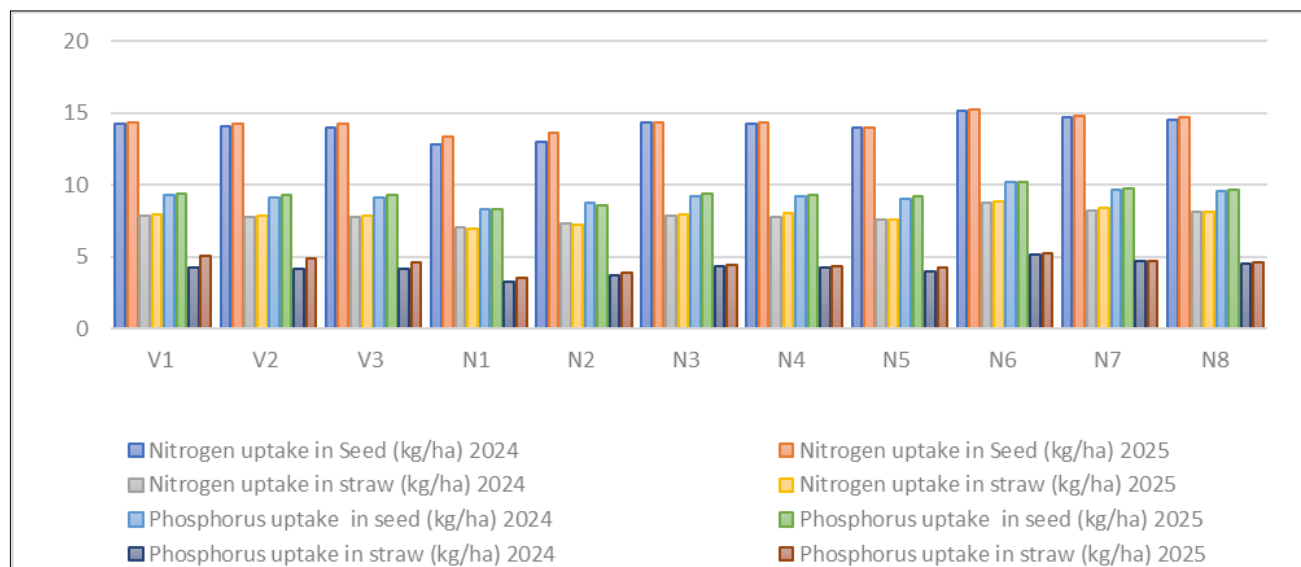


Fig. 2. Influence of integrated nutrient management (INM) and varieties on nitrogen and phosphorus uptake in seed and straw of mungbean (*Vigna radiata* L.). (DAP: diammonium phosphate; RDF: recommended dose of fertilisers).

Among INM treatments, T₆ (20 kg N + 60 kg P₂O₅ + *Rhizobium*) observed the increased N uptake in seed (15.15 and 15.24 kg ha⁻¹ during 2024 and 2025, respectively) and was found to be significantly higher than the remaining treatments. The lowest uptake was recorded under T₁ (15 kg N + 40 kg P₂O₅) (13.47 and 13.56 kg ha⁻¹). The enhanced uptake under T₆ is mainly due to combined application of higher N and P along with *Rhizobium* inoculation, which improves biological N fixation, root nodulation, nutrient availability and overall plant growth, resulting in greater nutrient translocation to seeds (29–31).

The highest N uptake in straw was observed in V₂ (MH 1142) (12.93 and 13.02 kg ha⁻¹ in 2024 and 2025), followed closely by V₁. The lowest uptake was recorded in V₃ (MH 1762) (11.84 and 11.92 kg ha⁻¹). Higher straw N uptake in V₂ indicates greater biomass production and better N assimilation in vegetative tissues. Treatment T₆ again recorded the maximum nitrogen uptake in straw (8.72 and 8.85 kg ha⁻¹ during 2024 and 2025), whereas T₁ resulted in the minimum uptake (7.04 and 7.12 kg ha⁻¹). The increase under T₆ can be attributed to synergistic effects of inorganic nutrients and biofertiliser, which promoted soil N availability and sustained nutrient release throughout crop growth.

The maximum P uptake in seed was recorded in V₂ (MH 1142) (15.24 and 15.29 kg ha⁻¹ in 2024 and 2025), while the minimum uptake occurred in V₃ (MH 1762) (14.15 and 14.50 kg ha⁻¹). This may be due to genotypic variation in root architecture and P use efficiency, enabling V₂ to absorb and translocate more P to seeds. Among INM treatments, T₆ recorded the highest P uptake in seed (10.16 and 10.19 kg ha⁻¹), followed by T₇ (RDF + nano urea + nano DAP). The lowest uptake was recorded under T₁ (8.48 and 8.56 kg ha⁻¹). Higher P uptake under T₆ is due to adequate P supply combined with *Rhizobium* activity, which enhances root growth and P solubilisation, thereby increasing P availability and uptake (32–34).

The pooled data for 2024 and 2025 in Table 5 revealed that a significant impact of varieties and INM treatments was observed on protein content in straw and seed of mungbean (Fig. 3). Among the varieties, V₁ (MH 421) resulted in maximum protein content in straw (13.07 and 13.12 % during 2024 and 2025, respectively), followed closely by V₂ (MH 1142), whereas the minimum straw protein content was observed in V₃ (MH 1762) (10.29 and 10.36 %). The differences were statistically significant as they exceeded the respective CD values. In contrast, seed protein content was highest in V₂ (MH 1142) with 20.12 and 20.22 % during 2024 and 2025, respectively, while the lowest values were recorded in V₃ (MH 1762) (13.33 and 13.48 %).

Table 5. Influence of integrated nutrient management and varieties on protein content in straw and seed of mungbean (*Vigna radiata* L.)

Treatments	Protein content in straw (%)		Protein content in seed (%)	
	2024	2025	2024	2025
Varieties				
V ₁ (MH 421)	13.07	13.12	19.39	19.46
V ₂ (MH 1142)	12.07	12.15	20.12	20.22
V ₃ (MH 1762)	10.29	10.36	13.33	13.48
SEm (±)	0.25	0.22	1.22	1.26
CD (at 5 %)	0.51	0.58	2.45	2.55
Integrated nutrient management				
T ₁ (15 kg N + 40 kg P ₂ O ₅)	4.96	4.99	7.11	7.26
T ₂ (15 kg N + 60 kg P ₂ O ₅)	6.64	6.72	7.92	7.96
T ₃ (20 kg N + 60 kg P ₂ O ₅)	8.16	8.25	12.16	12.21
T ₄ (15 kg N + 40 kg P ₂ O ₅ + <i>Rhizobium</i>)	6.84	6.92	10.40	10.65
T ₅ (15 kg N + 60 kg P ₂ O ₅ + <i>Rhizobium</i>)	8.22	8.26	13.11	13.45
T ₆ (20 kg N + 60 kg P ₂ O ₅ + <i>Rhizobium</i>)	9.12	9.24	14.22	14.36
T ₇ (RDF + 2 mL nano urea + 2 mL nano DAP ha ⁻¹)	7.42	7.65	11.15	11.33
T ₈ (RDF + 4 mL nano urea + 4 mL nano DAP ha ⁻¹)	7.67	7.72	12.02	12.12
SEm (±)	0.41	0.45	1.99	1.45
CD (at 5 %)	0.83	0.65	4.00	3.65

DAP: diammonium phosphate; RDF: recommended dose of fertilisers.

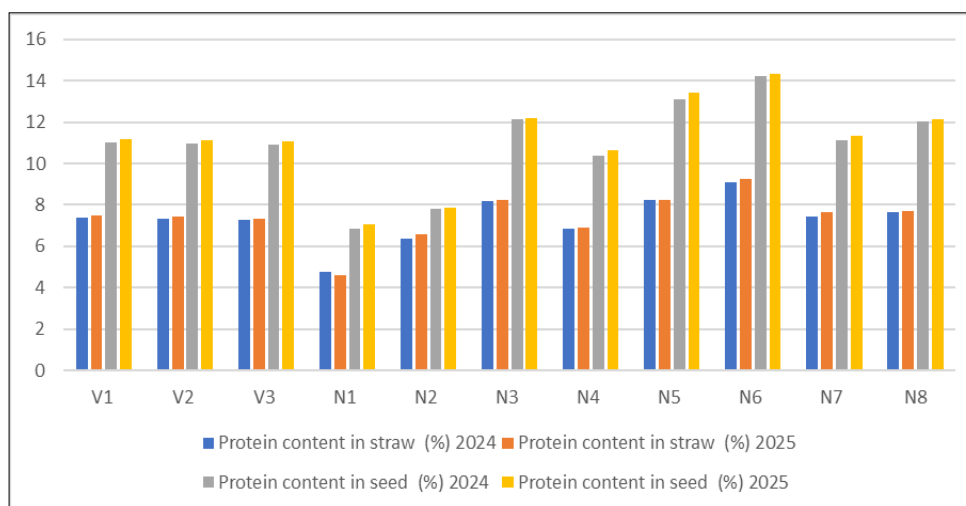


Fig. 3. Influence of integrated nutrient management (INM) and varieties on protein content in straw and seed of mungbean (*Vigna radiata* L.). (DAP: diammonium phosphate; RDF: recommended dose of fertilisers).

The superior seed protein content of MH 1142 may be attributed to its greater biological N fixation, higher nutrient uptake efficiency and more efficient translocation of nitrogenous compounds from vegetative tissues to developing seeds during the grain-filling stage. The comparatively higher protein concentration in the straw of MH 421 suggests differences in genotype-specific N partitioning, where a relatively larger proportion of assimilated N remained in vegetative tissues rather than being remobilised to the seeds. Conversely, the lower protein content in both straw and seed of MH 1762 may be associated with its comparatively lower nutrient acquisition efficiency, reduced N assimilation and genetic limitations in N remobilisation. These findings agree with previous reports that genotypic variation significantly influences N uptake, assimilation and partitioning, resulting in differences in seed protein content among mungbean varieties. However, unlike studies that reported a direct correspondence between high straw and seed protein content, the present investigation demonstrated differential N partitioning among varieties, with MH 421 accumulating more protein in vegetative tissues while MH 1142 exhibited more efficient remobilisation of N to the seeds (35–37). This indicates that varietal differences in protein content are governed not only by total N uptake but also by the efficiency of N utilisation and its redistribution during reproductive development.

A significant effect of INM practices were observed on protein concentration. Among the treatments, T_6 (20 kg N + 60 kg P_2O_5 + *Rhizobium*) recorded the maximum protein content in straw (9.12 and 9.24 %) and seed (14.22 and 14.36 %) during both years, whereas the minimum values were observed under T_1 (15 kg N + 40 kg P_2O_5), which recorded 4.96 and 4.99 % protein in straw and 7.11 and 7.26 % in seed during 2024 and 2025, respectively. The increase in protein content under T_6 could be attributed to the synergistic effect of greater N dose, adequate P supply and biological N fixation through *Rhizobium* inoculation, which enhanced N availability, amino acid synthesis and its translocation to seeds. Treatments involving *Rhizobium* (T_4 , T_5 and T_6) consistently performed better than inorganic fertilisation alone, highlighting the importance of INM in improving protein accumulation. Nano fertiliser treatments (T_7 and T_8) also improved protein content compared to lower fertiliser levels but remained inferior to the higher N and P combinations with *Rhizobium*. Overall, the results clearly indicate that balanced fertilisation along with biofertiliser inoculation plays a crucial role in enhancing protein content in both straw and seed of mungbean (38, 39).

Conclusion

The findings of the present study revealed that Integrated nutrient management (INM) significantly enhanced nutrient content and uptake in mungbean. MH 1142 recorded the highest nitrogen (N) and phosphorus (P) content as well as uptake in both seed and straw during both years, with N content of 5.28 and 5.32 % in seed, 3.32 and 3.36 % in straw, P content of 2.42 and 2.45 % in seed and 1.54 and 1.56 % in straw during 2024 and 2025, respectively. Similarly, it registered the maximum N uptake in seed (23.22 and 23.68 kg ha⁻¹) and straw (12.93 and 13.02 kg ha⁻¹) and P uptake in seed (15.24 and 15.29 kg ha⁻¹) and straw (7.25 and 7.29 kg ha⁻¹), indicating its superior nutrient acquisition and utilisation efficiency. Among the nutrient treatments, application of 20 kg N + 60 kg P_2O_5 along with *Rhizobium* application (T_6) proved most effective due to improved nutrient availability and biological N fixation. Nano-fertiliser treatments also showed beneficial effects but were comparatively less effective than integrated application of inorganic fertilisers and *Rhizobium*. Hence, cultivation of MH 1142 with 20 kg N + 60 kg P_2O_5 + *Rhizobium* is recommended for sustainable mung bean production under similar agro-climatic conditions.

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Authors' contributions

All authors have contributed equally to this research paper. All authors have read and approved the final manuscript.

Compliance with ethical standards

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Ethical issues: None

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used Claude to assist with improving the language, grammar, sentence structure and overall readability of the manuscript. After using this tool, the authors carefully reviewed and edited the content as needed and takes full responsibility for the publication.

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